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Use Case: Monitor Service Health & Alert on Downtime

Primary Actor: Scheduler (system timer) | **Secondary Actor:** DevOps Engineer (alert recipient)

Preconditions:

- The micro-service is registered in the catalog with a valid health-check endpoint URL.
- Health check monitoring is enabled for the service.
- An alerting channel (email/Slack) is configured for the subscribed DevOps Engineer.

Postconditions:

- The service's rolling 24-hour availability percentage is updated.
- The health check result is logged with a timestamp.
- If downtime was detected, an alert has been dispatched and the incident recorded.

Main Success Scenario:

1. Scheduler triggers a health check cycle every 30 seconds.
2. System retrieves the list of registered, active microservices and their endpoint URLs.
3. System sends an HTTP probe to each service's health endpoint.
4. Service responds with HTTP 200 within the timeout threshold.
5. System logs the successful probe result with a timestamp (*include: Log Health Check Result*).
6. System recomputes the rolling 24-hour availability percentage (*include: Compute Availability Percentage*).
7. System updates the dashboard to show the service as "Healthy."
8. Use case ends.

Alternate Flow — A1: Service Fails Health Probe (branches from step 3):

3a. Service does not respond, returns a non-200 status, or times out.

1. System logs the failed probe and increments the service's consecutive-failure counter.
2. If the counter is below 3, system marks the service "Degraded" and resumes at step 1 on the next cycle.
3. If the counter reaches 3, system marks the service "Down" and triggers the extension point → (*extend: Send Downtime Alert*) — composes and dispatches a notification to the subscribed DevOps Engineer containing service name, downtime start time, and last-healthy timestamp.
4. System records the downtime incident with a start timestamp for historical reporting.
5. Flow resumes at step 1; the incident stays open until a subsequent probe succeeds, at which point recovery time is logged and the incident is closed.

